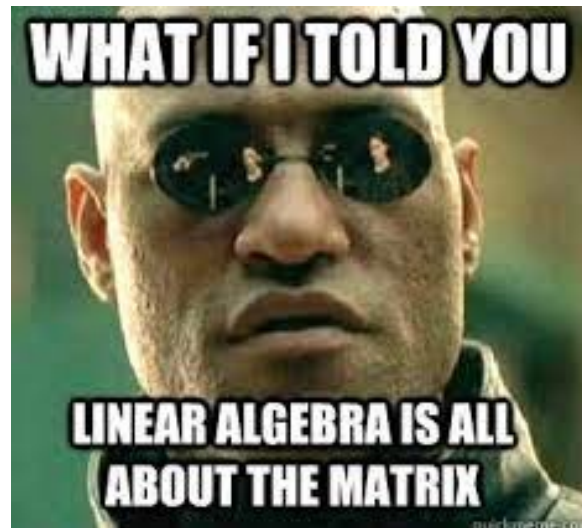


Matrix Algebra

Matrix (Linear) Algebra

- Why do you need to know (some) matrix algebra for this class?
 - The “data” are expressed in a (covariance) matrix and operated on from there
 - Some forms of SEM are describe in matrices (i.e., Bollen book)
 - Error messages from SEM programs will refer to specific matrices
 - Statistics generated from SEM only make sense with matrices (e.g., Model Fit)
- You are not expected to do matrix algebra “by hand”



Composition of a Matrix

$$\mathbf{A}_{3 \times 2} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$\mathbf{A}$ has 3 rows and 2 columns. It is a “3 by 2 matrix”

The information about the rows and columns is also called the **dimensions** of the matrix.

Individual entries in the matrix are called elements. If $\mathbf{A}$ has r rows and c columns it has $r \times c$ elements

Subscripts of the elements indicate their row and column (e.g., a_{32} is the element in the 3rd row and 2nd column)

A More General Notation

Matrix notation is *typically*...

$$\mathbf{A}_{p \times q}$$

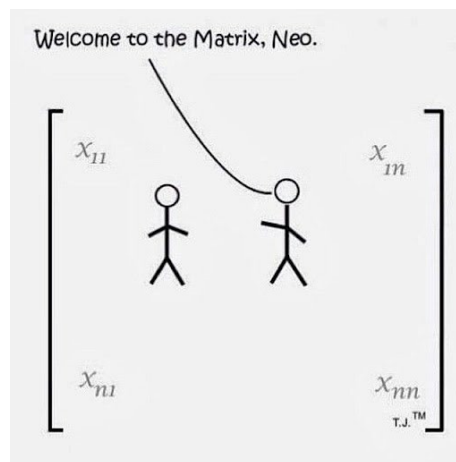
Matrix $\mathbf{A}$ is a capital and bold letter

It has p rows and q columns

a_{ij} is the element of $\mathbf{A}$ in row i and column j

One common exception....

Data matrices are typically described as $N \times p$ with N participants and p variables.



Matrix Transpose

A matrix transpose is used to swap rows and columns.

- The first row of A becomes the first column of A
- Second row of A becomes second column of A .
- etc.

$$A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \\ 9 & 8 \end{bmatrix} \quad A^T = A' = \begin{bmatrix} 1 & 5 & 9 \\ 3 & 7 & 8 \end{bmatrix}$$

Equivalent notation

A is 3×2 and A' is 2×3

An important property:

$$(A')' = A$$

Vectors

A column vector is a matrix with a single column (typically lower case bold)

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

A row vector is a matrix with a single row

$$\mathbf{a}' = [a_1 \quad a_2 \quad a_3]$$

For most of what we're doing, we'll typically assume a vector is a column vector unless otherwise noted. In which case, it is typically denoted as a transpose of a column vector

Adding and Subtracting Matrices

To add or subtract matrices they must be the same **dimensions**. Add or subtract the i, j th element from each matrix (element-wise operation)

$$\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 5 & 7 \\ 9 & 8 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 2 & 4 \\ 6 & 10 \\ -1 & -2 \end{bmatrix}$$

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} & \\ & \\ & \end{bmatrix} = \begin{bmatrix} & \\ & \\ & \end{bmatrix}$$

How do you think
you would solve
for $\mathbf{A} - \mathbf{B}$?

$$\mathbf{A} - \mathbf{B} = \begin{bmatrix} & \\ & \\ & \end{bmatrix} = \begin{bmatrix} & \\ & \\ & \end{bmatrix}$$

Important properties:

$$\begin{aligned} \mathbf{A} + \mathbf{B} &= \mathbf{B} + \mathbf{A} \\ \mathbf{A} - \mathbf{B} &\neq_n \mathbf{B} - \mathbf{A} \end{aligned}$$

Scalar Multiplication

You can multiply a matrix or a vector by a **scalar** (single number) by multiplying every element by that number.

$$c\mathbf{A} = c \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} = \begin{bmatrix} c \times a_{11} & c \times a_{12} \\ c \times a_{21} & c \times a_{22} \\ c \times a_{31} & c \times a_{32} \end{bmatrix}$$

If you want to **divide** every element by a scalar (c), you multiply by it's inverse ($1/c$)

$$\frac{1}{c}\mathbf{A} = \frac{1}{c} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} = \begin{bmatrix} \frac{a_{11}}{c} & \frac{a_{12}}{c} \\ \frac{a_{21}}{c} & \frac{a_{22}}{c} \\ \frac{a_{31}}{c} & \frac{a_{32}}{c} \end{bmatrix}$$

Scalar Multiplication

A special case worth considering is when $c = -1$

$$-1\mathbf{A} = -\mathbf{A} = -1 \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} = \begin{bmatrix} -a_{11} & -a_{12} \\ -a_{21} & -a_{22} \\ -a_{31} & -a_{32} \end{bmatrix}$$

Back to subtraction and addition for a moment:

Previously we noted that $\mathbf{A} - \mathbf{B} \neq_n \mathbf{B} - \mathbf{A}$

But now we have the vocabulary to give clearer statements:

$$\mathbf{A} - \mathbf{B} = -\mathbf{B} + \mathbf{A}$$

$$\mathbf{A} + (-\mathbf{B}) = \mathbf{A} - \mathbf{B}$$

Products with Vectors

Vectors must have **the same number of elements** to multiply

The **dot product** of two vectors (one row, one column vector) is the sum of the product of the elements of the two vectors.

$$\mathbf{x}' = [x_1 \ x_2 \ x_3] \quad \begin{matrix} y_1 x_1 \\ y_2 x_2 \\ y_3 x_3 \end{matrix} \quad \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \mathbf{y}$$

$$\begin{aligned} \mathbf{x}' \mathbf{y} &= y_1 x_1 + y_2 x_2 + y_3 x_3 \\ &= \sum_{i=1}^p x_i y_i \end{aligned}$$

Does this look similar to something we've already done?

Covariance is the dot product of two data vectors (divided by N) and

Variance is the dot product of a data vector with itself (divided by N)

Matrix Multiplication

Matrices that can be multiplied together are called **conformable***

The number of columns in the first matrix must equal the number of rows in the second matrix.

$$A_{r \times c} \times B_{c \times s} = (AB)_{r \times s}$$

These must match
conformable

Rows from first matrix
Columns from second matrix

***Conformable**: This is a common word to arise in error messages (e.g., “non-conformable arguments”). It is useful to know because it essentially means that something is the wrong size.

Matrix Multiplication

To find the elements of $\mathbf{AB}$ we take the **dot product** of the i^{th} row of $\mathbf{A}$ and the j^{th} column of $\mathbf{B}$

$$(ab)_{ij} = \mathbf{A}_{\text{Row}_i} \times \mathbf{B}_{\text{Col}_j} = \mathbf{a}'_i \mathbf{b}_j$$

Conformable?

$$\mathbf{B}_{2 \times 4} = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \end{bmatrix}$$

YES

$$3 \times \boxed{2} \text{ and } \boxed{2} \times 4$$

$$\mathbf{A}_{3 \times 2} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

What will the dimensions of $\mathbf{AB}$ be?

$$3 \times 4$$

$$(\mathbf{AB})_{3 \times 4} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} & a_{11}b_{13} + a_{12}b_{23} & a_{11}b_{14} + a_{12}b_{24} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} & a_{21}b_{13} + a_{22}b_{23} & a_{21}b_{14} + a_{22}b_{24} \\ a_{31}b_{11} + a_{32}b_{21} & a_{31}b_{12} + a_{32}b_{22} & a_{31}b_{13} + a_{32}b_{23} & a_{31}b_{14} + a_{32}b_{24} \end{bmatrix}$$

Some notes on Multiplication

Order matters (i.e., not **commutative**):

$$AB \neq_n BA$$

Premultiplication vs. Postmultiplication

AB then A is postmultiplied by B , or B is premultiplied by A

BA then B is postmultiplied by A , or A is premultiplied by B

Distributive Property

$$A(B + C) = AB + AC$$

$$(A + B)C = AC + BC$$

Transpose of a Product

$$(AB)' = B'A'$$

Types of Matrices

Square

Same number of rows and columns

Covariance matrix and correlation matrix are square matrices

Square matrices can be multiplied by themselves (e.g., A^2 , A^3)

$$A_{3 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Symmetric

Must be **square**

Entries above and below the diagonal are the same (i.e., $a_{ij} = a_{ji}$)

$$A_{3 \times 3} = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{21} & a_{22} & a_{32} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Type of Matrices

Diagonal

Must be square and symmetric

All off diagonal values are zero

Diagonal elements should **not all** be zero

$$\mathbf{A}_{3 \times 3} = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$$

Demonstrate Conversion of Σ to P

$$P = D\Sigma D$$

where D is a square matrix with the inverses of the standard deviations on the diagonals

$$P = \begin{bmatrix} 1 & \rho_{xy} & \rho_{xz} \\ \rho_{xy} & 1 & \rho_{yz} \\ \rho_{xz} & \rho_{yz} & 1 \end{bmatrix} = \begin{bmatrix} 1 & \frac{\sigma_{xy}}{\sigma_x \sigma_y} & \frac{\sigma_{xz}}{\sigma_x \sigma_z} \\ \frac{\sigma_{xy}}{\sigma_x \sigma_z} & 1 & \frac{\sigma_{yz}}{\sigma_y \sigma_z} \\ \frac{\sigma_{xz}}{\sigma_x \sigma_z} & \frac{\sigma_{yz}}{\sigma_y \sigma_z} & 1 \end{bmatrix}$$

Insight: Premultiplying by a diagonal matrix distributes values along rows, postmultiplying distributes values along columns

$$D = \begin{bmatrix} 1/\sigma_x & 0 & 0 \\ 0 & 1/\sigma_y & 0 \\ 0 & 0 & 1/\sigma_z \end{bmatrix}$$

$$D\Sigma = \begin{bmatrix} \sigma_x^2/\sigma_x & \sigma_{xy}/\sigma_x & \sigma_{xz}/\sigma_x \\ \sigma_{xy}/\sigma_y & \sigma_y^2/\sigma_y & \sigma_{yz}/\sigma_y \\ \sigma_{xz}/\sigma_z & \sigma_{yz}/\sigma_z & \sigma_z^2/\sigma_z \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_y^2 & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_z^2 \end{bmatrix}$$

$$D\Sigma D = \begin{bmatrix} \sigma_x^2/\sigma_x\sigma_x & \sigma_{xy}/\sigma_x & \sigma_{xz}/\sigma_x\sigma_z \\ \sigma_{xy}/\sigma_y\sigma_x & \sigma_y^2/\sigma_y & \sigma_{yz}/\sigma_y\sigma_z \\ \sigma_{xz}/\sigma_z\sigma_x & \sigma_{yz}/\sigma_z & \sigma_z^2/\sigma_z\sigma_z \end{bmatrix}$$

Identity Matrix

- Square matrix with all 1s on diagonal and 0s on off diagonal

$$\mathbf{I}_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- $\mathbf{I}$ is special and reserved in matrix algebra
- It is equivalent to 1 in scalar algebra

Assume $\mathbf{I}$ is conformable with $\mathbf{A}$ and $\mathbf{B}$

$$\mathbf{A} \mathbf{I} = \mathbf{A}$$

$$\mathbf{I} \mathbf{B} = \mathbf{B}$$

Remember how $X^0 = 1$? For a square matrix $\mathbf{A}$

$$\mathbf{A}_{p \times p}^0 = \mathbf{I}_{p \times p}$$

Zero Matrix

Sometimes you need a matrix of all zeros

$$\mathbf{0}_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

All entries are zero

Typically square (doesn't have to be)

Any conformable matrix pre- or post-multiplied by $\mathbf{0}$ is $\mathbf{0}$ of the appropriate dimensions

$$\mathbf{A}_{3 \times 2} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

What will the dimensions of the result be?

$$\mathbf{A}_{3 \times 2} \mathbf{0}_{2 \times 2} =$$

Trace of a Matrix

The sum of the diagonal of a square matrix is its **trace**

$$tr(\mathbf{I}_{3 \times 3}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$tr(\mathbf{A}_{3 \times 3}) = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{21} & a_{22} & a_{32} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$tr(\mathbf{\Sigma}) = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_y^2 & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_z^2 \end{bmatrix}$$

Trace of a non-square matrix is not defined

Partitioned Matrices

Matrices can be broken up into pieces

This is useful because this can help apply common properties to specific pieces.

$$\mathbf{F}_{2 \times 4} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$\mathbf{F}$ doesn't have a special name or known properties, but...

$$\mathbf{F}_{2 \times 4} = \begin{bmatrix} 1 & 0 & | & 0 & 0 \\ 0 & 1 & | & 0 & 0 \end{bmatrix}$$

$\mathbf{F}$ is made up of $\mathbf{I}_{2 \times 2}$ and $\mathbf{0}_{2 \times 2}$

$$\mathbf{F}_{2 \times 4} = \left[\mathbf{I}_{2 \times 2} \quad \mathbf{0}_{2 \times 2} \right]$$

Partitioned Matrices

Consider a matrix $A_{4 \times 4}$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$$

$$A_{1_{2 \times 4}} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \end{bmatrix}$$

$$A_{2_{2 \times 4}} = \begin{bmatrix} a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

then

$$F_{2 \times 4} A_{4 \times 4} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \end{bmatrix}$$

$$= \begin{bmatrix} I_{2 \times 2} & \mathbf{0}_{2 \times 2} \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = A_1$$

Partitioning is more of a mathematical tool than a property of a matrix

Matrix Inverses

In scalar algebra, dividing by a number is the same as multiplying by the inverse.

$$\frac{a}{b} = a \times \frac{1}{b}$$

Also a number multiplied by it's inverse is 1

$$a \times \frac{1}{a} = 1$$

In matrix algebra there is no “matrix division” only inverses
 $\mathbf{AB}^{-1}$

and the idea behind an inverse is

$$\mathbf{A}_{p \times p} \mathbf{A}_{p \times p}^{-1} = \mathbf{I}_{p \times p}$$

Inverses only exist for square matrices

Taking an Inverse

I will never ask you to do this by hand in this class but... just so you know

$$A_{2 \times 2} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

change the sign in
the off diagonal
swap the diagonal

This is called the **determinant**

For anything larger than 2 x 2 use a computer
`solve(A)` in R

Does the inverse exist?

As long as the determinant is not zero, the inverse exists
`det(A)` in R

Matrices with 0 determinants are called **singular**

Matrices with non-zero determinants are called **non-singular**

Rank of a Matrix

Formal Definition: Dimension of the vector space spanned by column vectors

Practical Definition: Number of linearly independent columns

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \qquad \text{rank}(\mathbf{A}) = 3$$

Because the 4th column is the sum of the 1st, 2nd, and 3rd it does not add any dimensionality

$$\mathbf{B} = \begin{bmatrix} 1 & 2 & 8 & 0 \\ 2 & 4 & 16 & 0 \\ 3 & 6 & 24 & 0 \end{bmatrix} \qquad \text{rank}(\mathbf{B}) = 1$$

2nd column is 1st column time 2, 3rd column is 1st column times 8 (or 2nd column times 4), and 4th column is any column times 0

Rank is another term that arises in error messages, (e.g., “rank deficient” or “singular matrix”). Matrices of deficient rank often cannot be inverted.

Summary

Matrices are the primary object of analysis in structural equation modeling

Understanding matrix algebra (even if you don't do it by hand) will help you:

- Understand your models better
- Understand and debug error messages from software
- Use specific software that relies on matrices
- Understand the output of your models

Core concepts:

- Matrices & Vectors
- Dimension & Rank
- Transpose
- Adding and multiplying
- Inverses and Traces
- Square, Symmetric, and Diagonal Matrices
- Partitions
- Identity and Zero matrices